

Changelog:

V4.0 Date 3.27.25

- Added "Mix & Match" Field Maps
- Updated Calendar Pages

V3.0 Date 6.10.24

- Added "Rapid Relay" Field Map
- Updated Calendar pages

V2.1 Date 6.12.23

- Added new calendar and planner layouts starting on slide 21. These are *not* in the "layouts" dropdown, so if you delete one by mistake you will need to copy it from the template again. This keeps the layout options from getting too crowded.
- Added more customizable front covers. There is an option with black text for lighter colors and an option with white text for darker colors.
- Moved extra controller map and brain map assets to parts library document
- Changed the divider page to be customizable.

V2.0 Date 5.16.23

- Added second title page template for additional customizability
- Added new template pages for controller map & brain map
- Added "Full Volume" Field Maps

**You can delete this page
from your Team's
notebook!**

This is the basic notebook cover page.

Engineering Notebook

0000

Team Number

Your Team Name Here

Team Name

Your School Name Here

School

00/00/0000

Start Date

00/00/0000

End Date

1

Book #

of 1

V4.0 Date 3.27.25



Engineering Notebook

This is an alternate front page.
Place a team photo or other information here.

0000

Team Number

Your Team Name Here

Team Name

Your School Name Here

School

00/00/0000

Start Date

00/00/0000

End Date

1

Book #

of 1

V4.0 Date 3.27.25



Engineering Notebook

This is an alternate front page.
Place a team photo or other information here.

0000

Team Number

Your Team Name Here

Team Name

Your School Name Here

School

00/00/0000

Start Date

00/00/0000

End Date

1

Book #

of 1

V4.0 Date 3.27.25



Engineering Notebook

This is an alternate front page.
Place a team photo or other information here.

0000

Team Number

Your Team Name Here

Team Name

Your School Name Here

School

00/00/0000

Start Date

00/00/0000

End Date

1

Book #

of 1

V4.0 Date 3.27.25

Resources

students.vex.com

Engineering Resources, Information on Notebooks, Videos, VEX Library, Teams Resources, and Scholarships



mentors.vex.com

Team and Mentor Resources, Mentor Professional Development, VEX Mentor Community and more



teams.vex.com

A Collection of Resources for Teams Provided by the REC Foundation



library.vex.com

Information on Building, Documentation, Troubleshooting, Coding, and other Educational Resources



About the REC Foundation

The REC Foundation's global mission is to provide educators with hands-on, student-led competition programs and educational resources to prepare future innovators for a diverse and inclusive STEM workforce. We see a future where all students design and innovate as part of a team, experience failure, persevere, and emerge confident in their ability to meet global challenges.

engineering.vex.com

notebooking.vex.com

coding.vex.com

Judging Rubric for Notebooks

vex.com

roboticseducation.org

VIQRC 2025-2026 Game - Rules & Game Video

Send suggestions and comments about these Digital Notebooks and Digital Parts to **notebooks@vex.com**

Table of Contents

Project

Name

Date _____

Page

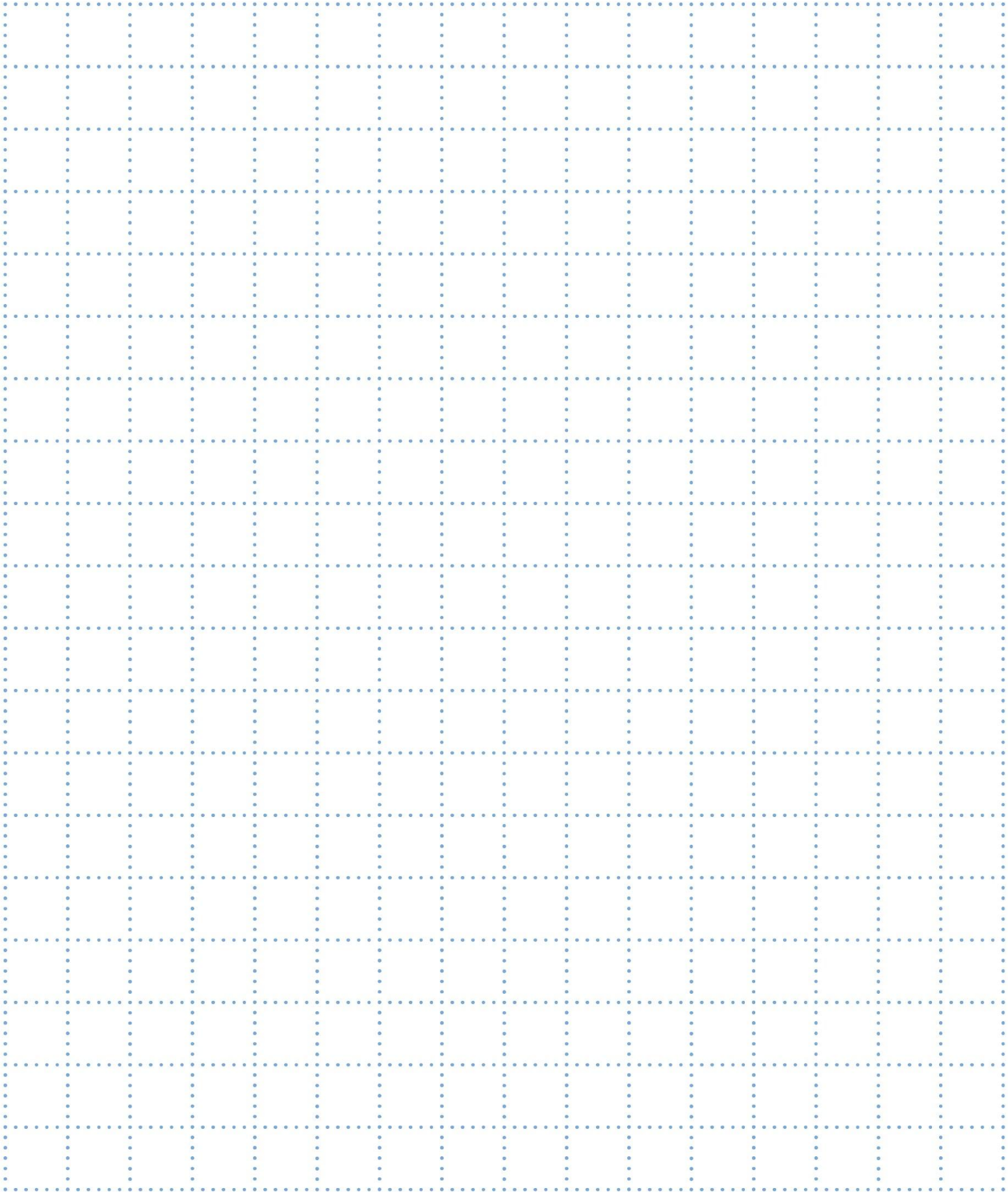
Project

Name

Date

Page

Section A

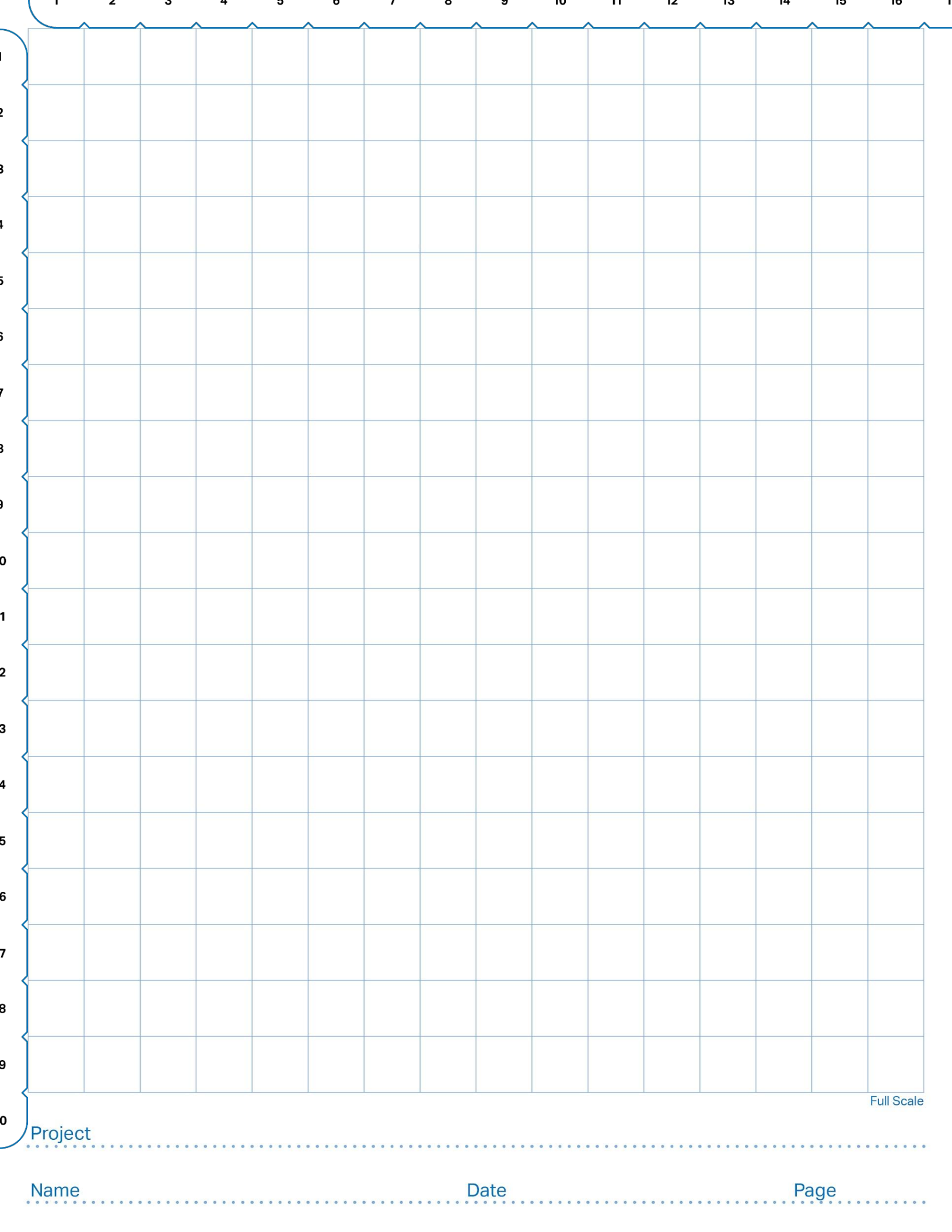


Project

Name

Date

Page



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1																				
2																				
3																				
4																				
5																				
6																				
7																				
8																				
9																				
10																				
11																				
12																				
13																				
14																				
15																				
16																				
17																				
18																				
19																				
20																				

Half Scale

Project

Name Date Page

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1																				
2																				
3																				
4																				
5																				
6																				
7																				
8																				
9																				
10																				
11																				
12																				
13																				
14																				
15																				
16																				
17																				
18																				
19																				
20																				

Half Scale

Project

Name Date Page

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1																				
2																				
3																				
4																				
5																				
6																				
7																				
8																				
9																				
10																				
11																				
12																				
13																				
14																				
15																				
16																				
17																				
18																				
19																				
20																				

Half Scale

Project

Name Date Page

[illegible]

Project

Name _____ Date _____ Page _____

Port 7

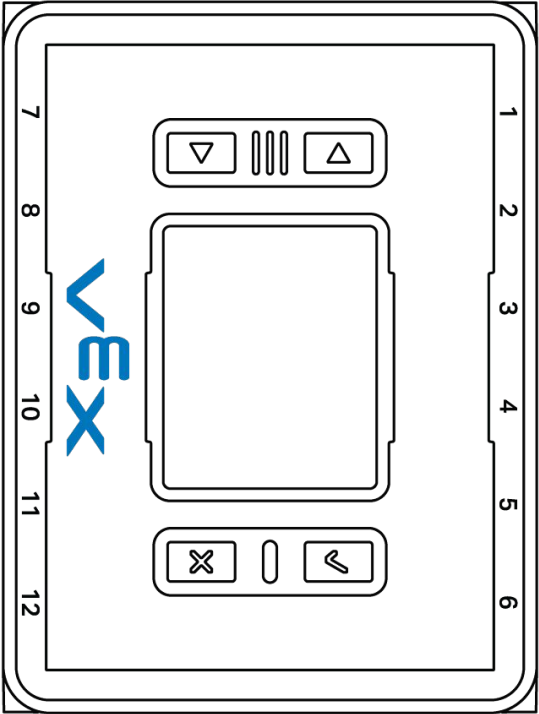
Port 8

Port 9

Port 10

Port 11

Port 12



1 Port

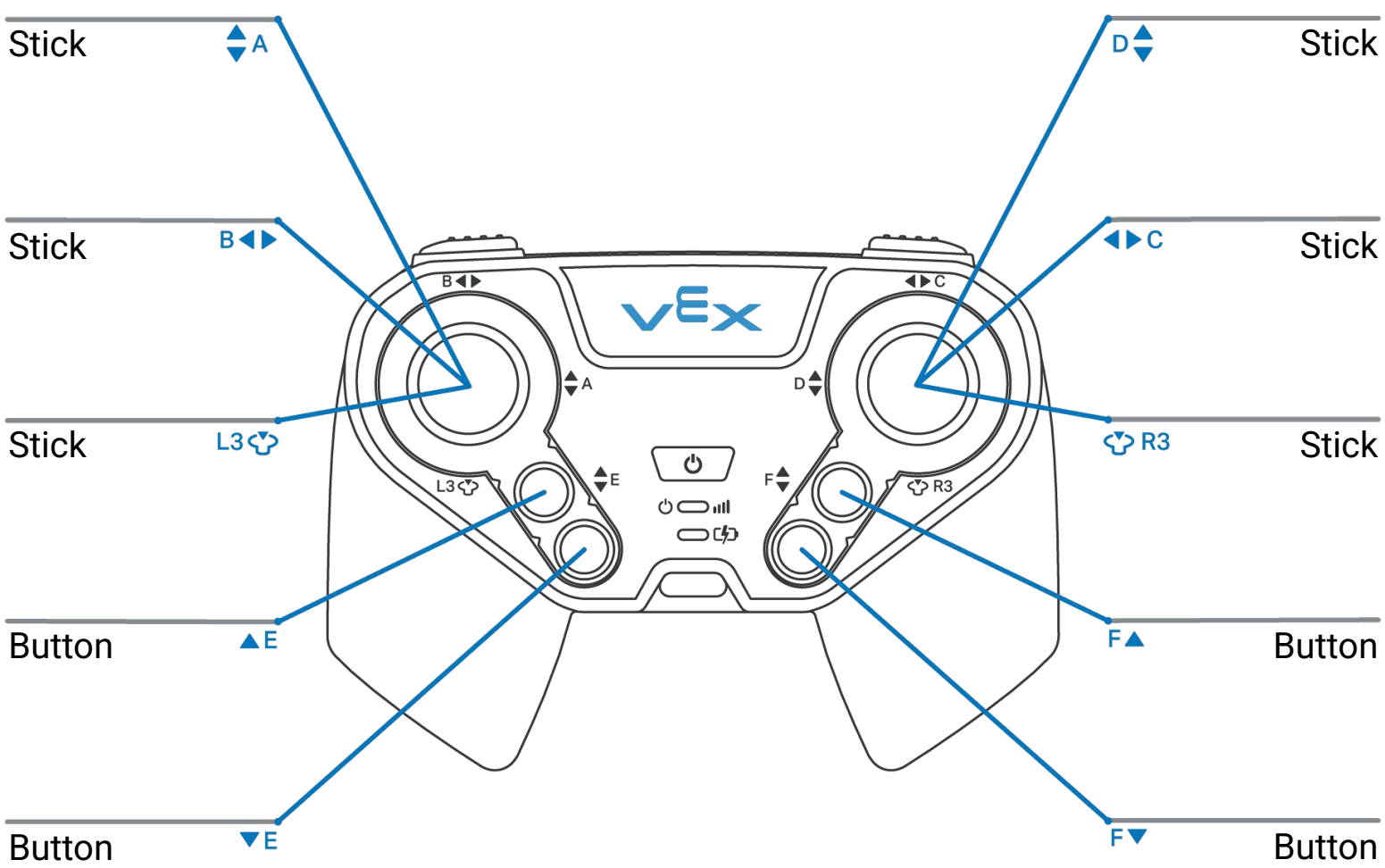
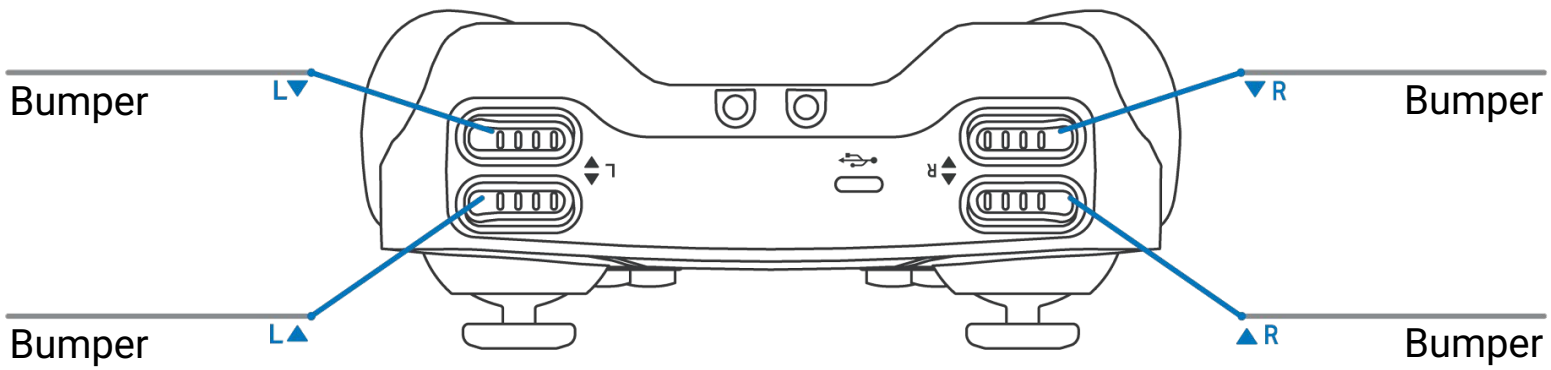
2 Port

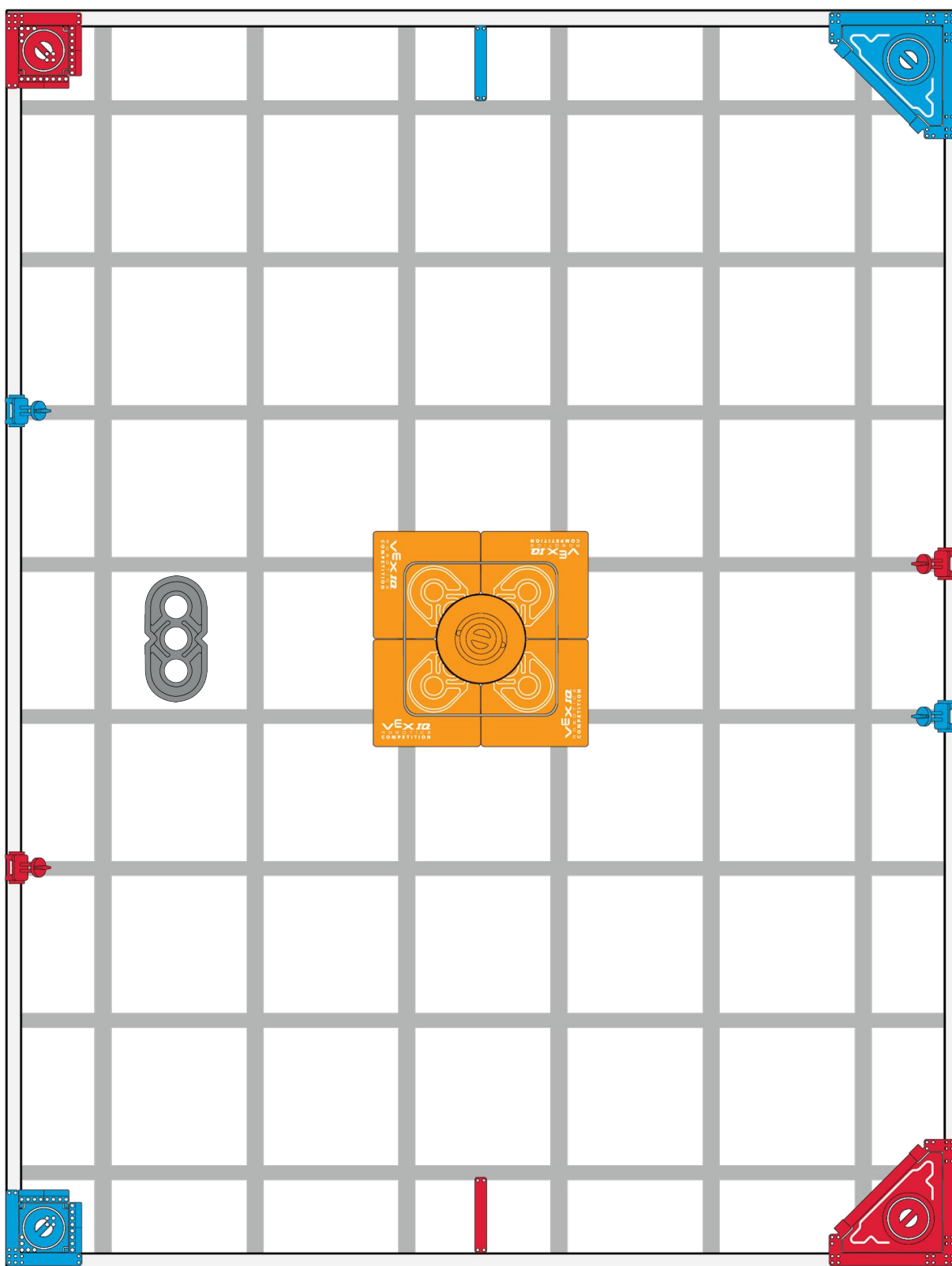
3 Port

4 Port

5 Port

6 Port



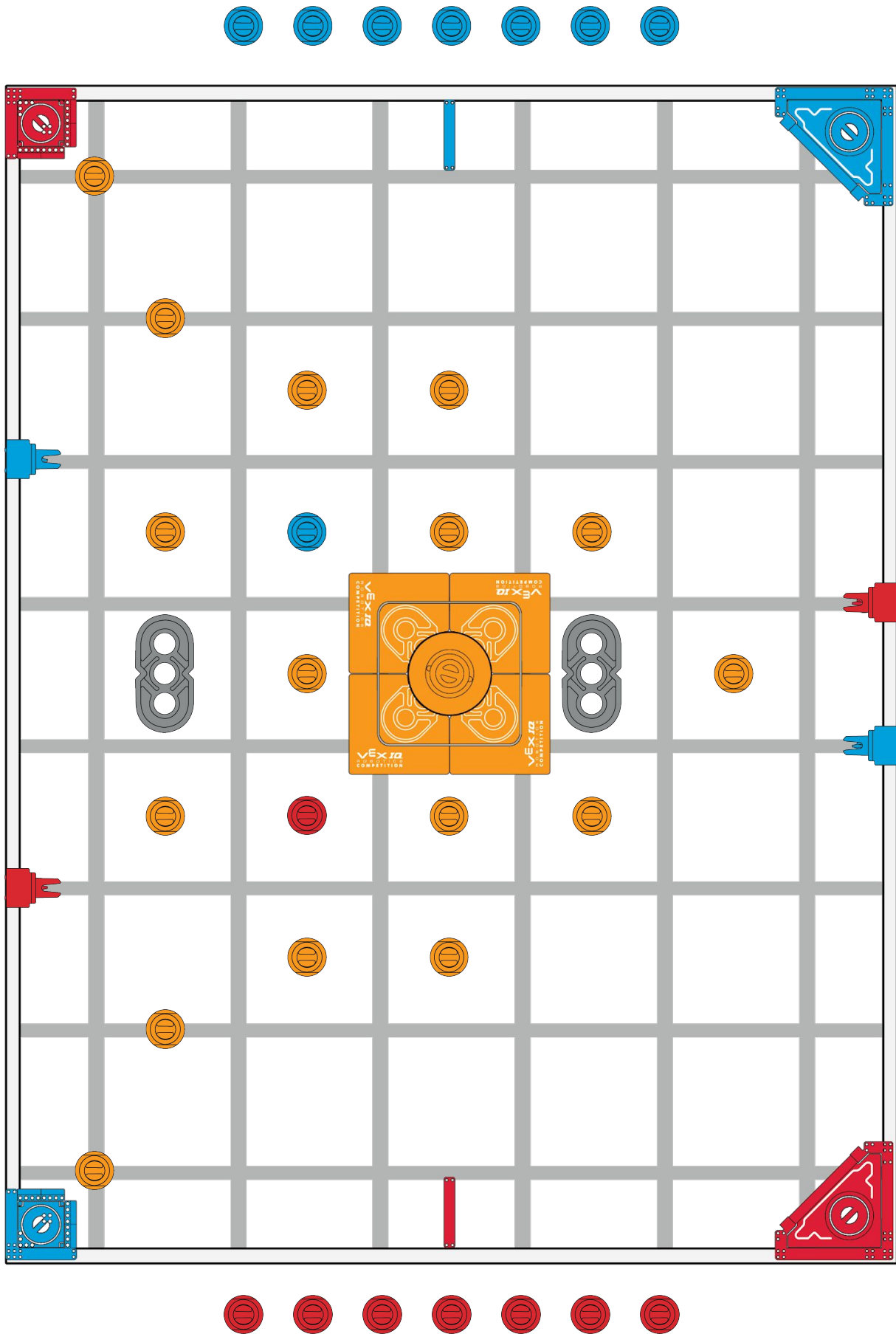


Project "Mix & Match" Blank Field

Name

Date

Page

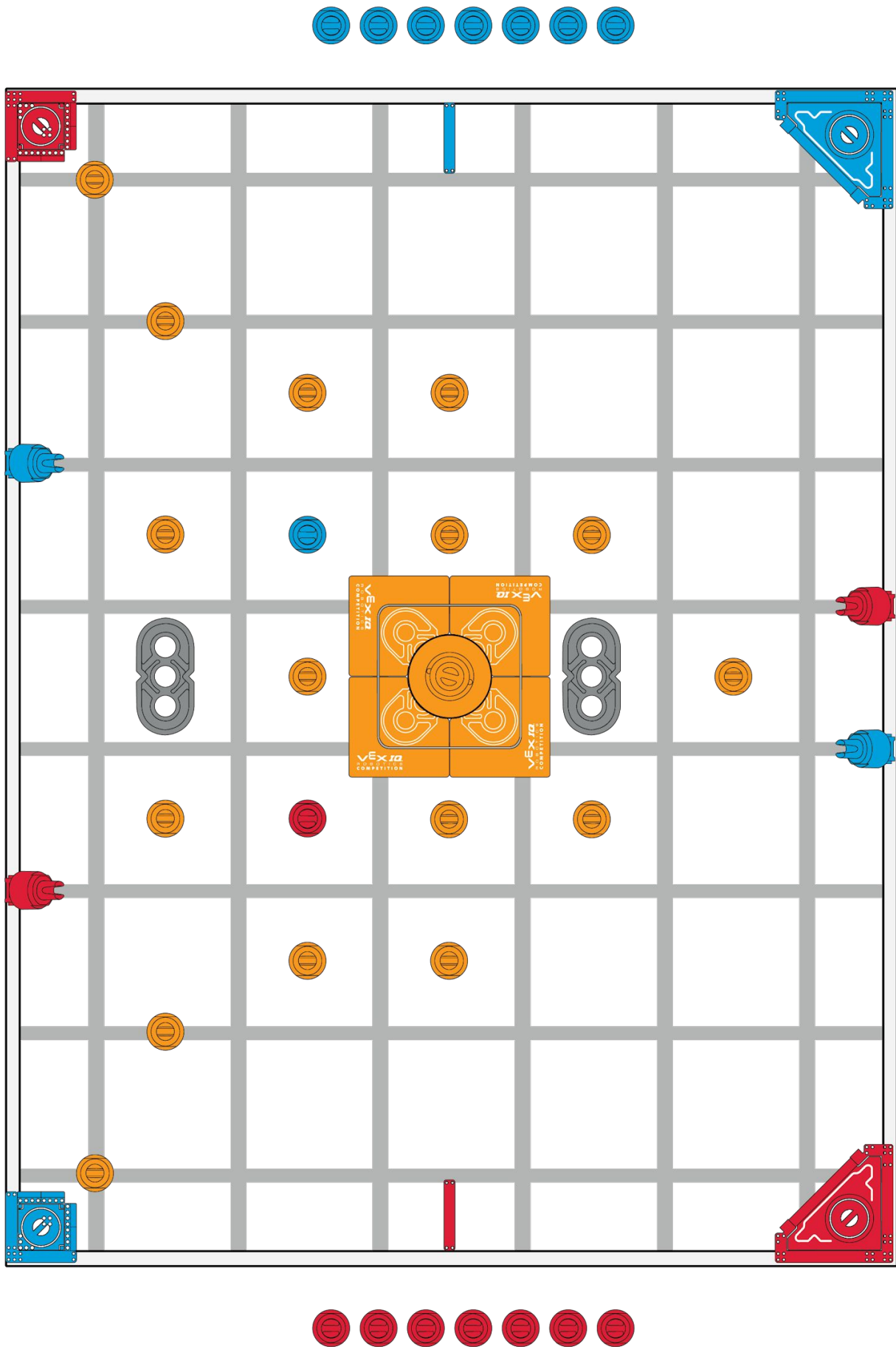


Project "Mix & Match" Starting Field with Moveable Game Objects

Name

Date

Page



Project "Mix & Match" Starting Field with Static Game Objects

Name

Date

Page

Gear Formulas

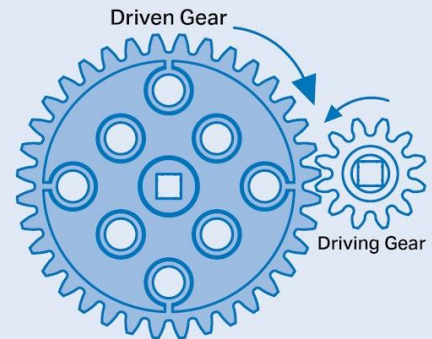
$$\text{Gear Ratio} = \frac{\text{\# of Driven Gear Teeth (Output)}}{\text{\# of Driving Gear Teeth (Input)}}$$

Power Transfer is a 1:1 gear ratio where the driving and driven gear have the **same number** of teeth.

Increasing Torque (lowering speed) is a gear ratio where the driving gear has **fewer teeth** than the driven gear.

Increasing Speed (lowering torque) is a gear ratio where the driving gear has **more teeth** than the driven gear.

$$\text{Compound Gear Ratio} = (\text{Gear Ratio 1}) \times (\text{Gear Ratio 2}) \times (...)$$



Motion Formulas

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

Distance is from the axis of rotation

$$\text{Rotational Speed} = \frac{\text{\# of Turns}}{\text{Time}} = \frac{\text{Degrees}}{\text{Time}}$$

$$\text{Circumference} = \pi \times \text{Diameter}$$

$$\text{Power} = \text{Force} \times \text{Velocity}$$

$$\pi \approx 3.14$$

$$\text{Torque} = \text{Force} \times \text{Distance}$$

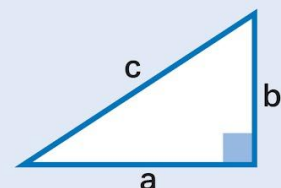
$$\text{Force} = \text{Mass} \times \text{Acceleration}$$

Mathematical Formulas

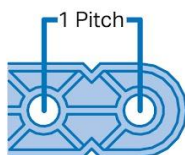
Complimentary angles are angles that sum to 90°

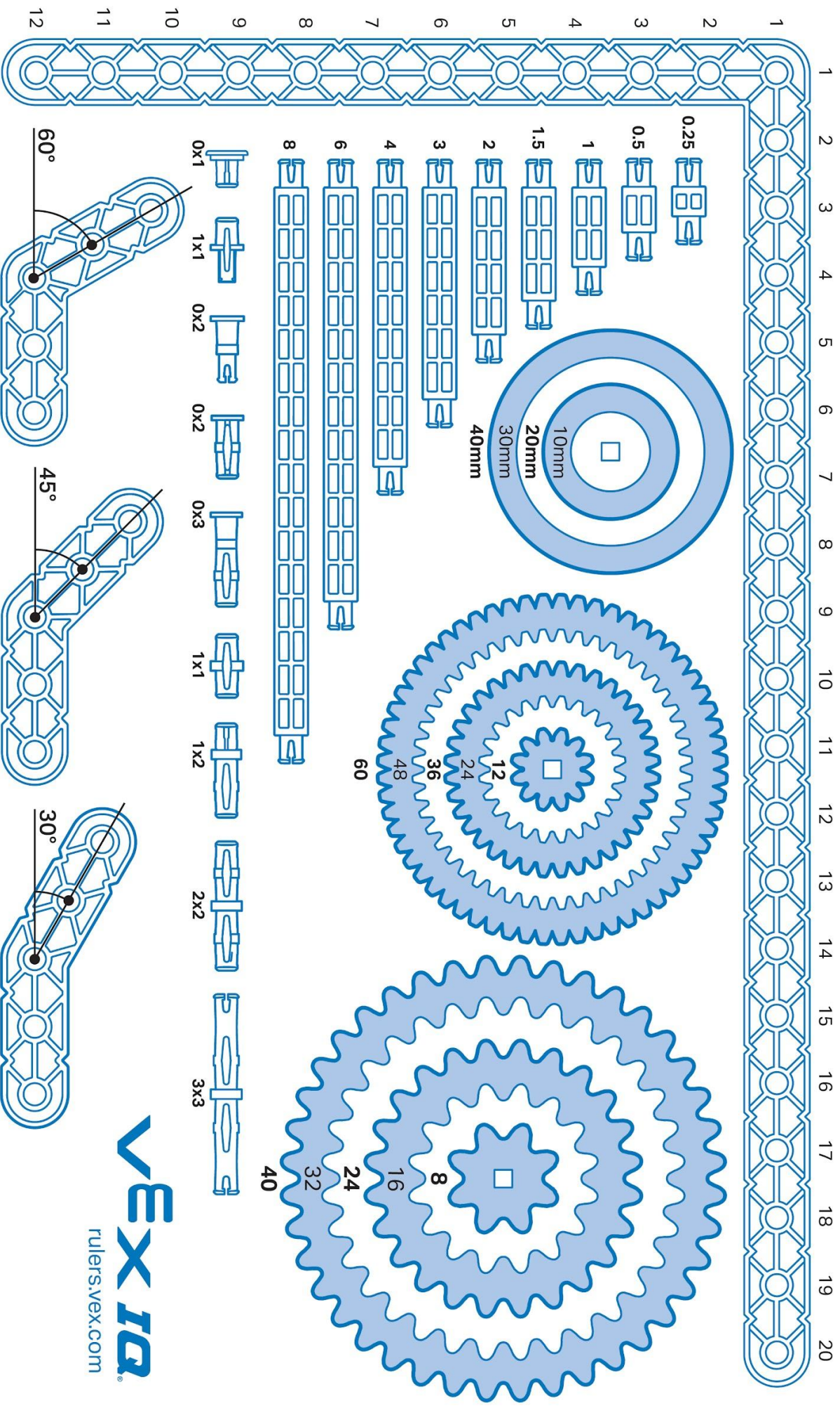
Supplementary angles are angles that sum to 180°

$$\text{Pythagorean Theorem: } c^2 = a^2 + b^2$$



$$1 \text{ Pitch} = 0.5 \text{ in} = 12.7 \text{ mm}$$





Week of: MM/DD/YYYY

Monday	
Tuesday	
Wednesday	
Thursday	
Friday	
Saturday	Sunday

May 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

June 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

July 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

August 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

September 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

October 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

November 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

December 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

January 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

February 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28

March 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

April 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

May 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

May 2025

T	1
F	2
S	3
S	4
M	5
T	6
W	7
T	8
F	9
S	10
S	11
M	12
T	13
W	14
T	15
F	16
S	17
S	18
M	19
T	20
W	21
T	22
F	23
S	24
S	25
M	26
T	27
W	28
T	29
F	30
S	31

June 2025

S	1
M	2
T	3
W	4
T	5
F	6
S	7
S	8
M	9
T	10
W	11
T	12
F	13
S	14
S	15
M	16
T	17
W	18
T	19
F	20
S	21
S	22
M	23
T	24
W	25
T	26
F	27
S	28
S	29
M	30

July 2025

T 1

W 2

T 3

F 4

S 5

S 6

M 7

T 8

W 9

T 10

F 11

S 12

S 13

M 14

T 15

W 16

T 17

F 18

S 19

S 20

M 21

T 22

W 23

T 24

F 25

S 26

S 27

M 28

T 29

W 30

T 31

August 2025

F 1

S 2

S 3

M 4

T 5

W 6

T 7

F 8

S 9

S 10

M 11

T 12

W 13

T 14

F 15

S 16

S 17

M 18

T 19

W 20

T 21

F 22

S 23

S 24

M 25

T 26

W 27

T 28

F 29

S 30

S 31

September 2025

M	1
T	2
W	3
T	4
F	5
S	6
S	7
M	8
T	9
W	10
T	11
F	12
S	13
S	14
M	15
T	16
W	17
T	18
F	19
S	20
S	21
M	22
T	23
W	24
T	25
F	26
S	27
S	28
M	29
T	30

October 2025

W 1

T 2

F 3

S 4

S 5

M 6

T 7

W 8

T 9

F 10

S 11

S 12

M 13

T 14

W 15

T 16

F 17

S 18

S 19

M 20

T 21

W 22

T 23

F 24

S 25

S 26

M 27

T 28

W 29

T 30

F 31

November 2025

S 1

S 2

M 3

T 4

W 5

T 6

F 7

S 8

S 9

M 10

T 11

W 12

T 13

F 14

S 15

S 16

M 17

T 18

W 19

T 20

F 21

S 22

S 23

M 24

T 25

W 26

T 27

F 28

S 29

S 30

December 2025

M 1

T 2

W 3

T 4

F 5

S 6

S 7

M 8

T 9

W 10

T 11

F 12

S 13

S 14

M 15

T 16

W 17

T 18

F 19

S 20

S 21

M 22

T 23

W 24

T 25

F 26

S 27

S 28

M 29

T 30

W 31

January 2026

T 1

F 2

S 3

S 4

M 5

T 6

W 7

T 8

F 9

S 10

S 11

M 12

T 13

W 14

T 15

F 16

S 17

S 18

M 19

T 20

W 21

T 22

F 23

S 24

S 25

M 26

T 27

W 28

T 29

F 30

S 31

February 2026

S	1
M	2
T	3
W	4
T	5
F	6
S	7
S	8
M	9
T	10
W	11
T	12
F	13
S	14
S	15
M	16
T	17
W	18
T	19
F	20
S	21
S	22
M	23
T	24
W	25
T	26
F	27
S	28

March 2026

S 1
M 2
T 3
W 4
T 5
F 6
S 7
S 8
M 9
T 10
W 11
T 12
F 13
S 14
S 15
M 16
T 17
W 18
T 19
F 20
S 21
S 22
M 23
T 24
W 25
T 26
F 27
S 28
S 29
M 30
T 31

April 2026

W 1

T 2

F 3

S 4

S 5

M 6

T 7

W 8

T 9

F 10

S 11

S 12

M 13

T 14

W 15

T 16

F 17

S 18

S 19

M 20

T 21

W 22

T 23

F 24

S 25

S 26

M 27

T 28

W 29

T 30

May 2026

F	1
S	2
S	3
M	4
T	5
W	6
T	7
F	8
S	9
S	10
M	11
T	12
W	13
T	14
F	15
S	16
S	17
M	18
T	19
W	20
T	21
F	22
S	23
S	24
M	25
T	26
W	27
T	28
F	29
S	30
S	31

